

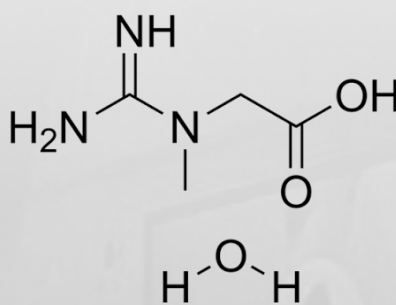
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CREATINE MONOHYDRATE

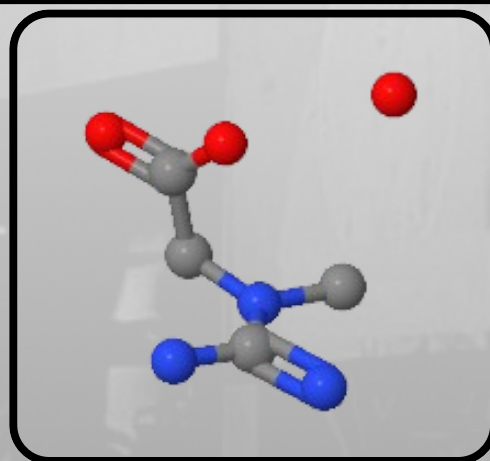
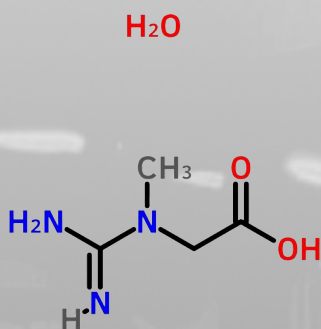
Adenosine Triphospate (ATP) Regenerator

CHEMICAL IDENTITY

IUPAC name: 2-[carbamimidoyl(methyl)amino]
acetic acid; hydrate
Molecular formula: $C_4H_9N_3O_2 \cdot H_2O$
Equivalent combined formula: $C_4H_{11}N_3O_3$
Molar mass: 149.15 g/mol
CAS No.: 6020-87-7
PubChem CID: 80116



STRUCTURE (Creatine Monohydrate)

INTERESTING
FACT

Creatine is stored in muscle partly as phosphocreatine. During short periods of high energy demand, phosphocreatine can donate a phosphate group to ADP, helping rapidly regenerate ATP—the cell's immediate energy currency.

KEY PROPERTIES & DATA

Appearance: White crystalline solid/powder
Molar mass: 149.15 g/mol
Water solubility: Limited compared with highly soluble small molecules; solubility is temperature-dependent
Hydrogen-bond donors: 4
Hydrogen-bond acceptors: 4
Topological polar surface area: 91.4 Å²
Form: Hydrated crystalline compound

USES & VALUE

Biological role

- Naturally occurring compound involved in cellular energy metabolism.
- Creatine is present in vertebrate tissues and is particularly associated with muscle metabolism.

Commercial/consumer use

- Creatine monohydrate is widely used as an ingredient in nutritional products.



SOURCES

Red meat, Fish (Tuna, Salmon, Milk Fish),
Eggs, Supplements (Powdered Creatine)

RISKS AND
SAFETY

- Generally well tolerated when used appropriately.
- Gastrointestinal discomfort can occur in some people.
- Because creatine is associated with water storage in muscle, changes in body water can occur.
- People with existing medical conditions should seek appropriate professional advice before using supplements.
- Use reputable products and follow the product's directions.

CHEMICAL ENGINEERING
CONNECTION

- Reaction engineering — controlled synthesis
- Crystallization — recovery of the solid product
- Filtration/separation — removal of mother liquor and impurities
- Drying — control of moisture/hydrate form
- Particle-size control — important for powder processing
- Quality control — purity, identity, moisture, and product consistency
- Formulation — incorporation into finished products

